

ROBERTO VILLAFUERTE

AI/ML Engineer

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PROFESSIONAL PROFILE

Computer Science final-year undergraduate specializing in MLOps and RAG systems. Experience architecting automated forecasting pipelines (Snowflake/Mage AI) and deploying multi-tenant GenAI support agents. Completed Master's-level coursework in Trustworthy ML at the University of Helsinki. Focused on shipping scalable, bias-aware production models.

EXPERIENCE

COMPUMAX <i>AI Engineer</i>	Remote, Ecuador Dec 2025 – Present
- Architecting a multi-tenant RAG helpdesk for small businesses; reduced support ticket resolution time by roughly 30% (projected) by automating responses using proprietary vendor data. - Designing internal Generative AI workflows to modernize operations, ensuring alignment between technical architecture and unit economics.	
Datalysis Group <i>Machine Learning Engineer Intern</i>	Remote, Ecuador Feb 2025 – Jun 2025
- Built an automated demand forecasting pipeline for inventory replenishment using Snowflake and Mage AI, ingesting daily transaction data to support restocking decisions for operations and finance. - Implemented an end-to-end ML workflow, automating data cleaning/validation, feature engineering, training, and evaluation for reliable production runs.	

TECHNICAL PROJECTS

Fairness-Aware Candidate Pre-Screening <i>Python, Docker, CI/CD, Scikit-learn</i>	Jan 2026
- Developed a production-grade ML pipeline for automated hiring that includes bias mitigation via Equal Opportunity post-processing. - Integrated Fairness Metrics (SPD, DI, TPR Gap) directly into the evaluation loop to prevent algorithmic discrimination. - Engineered a robust deployment environment using Docker and established CI/CD workflows for consistent testing.	
Seismic Multi-Label Event Classifier <i>Python, Signal Processing, Data Science</i>	
- Architected an end-to-end data science pipeline to classify complex seismic signals (.at2 format) using multi-label classification techniques. - Optimized feature extraction from raw acoustic data to improve event detection accuracy in noisy environments.	

EDUCATION

University of Helsinki <i>Exchange Studies, Master's Level Data Science (GPA: 4.83/5.0)</i>	Helsinki, Finland Aug 2025 – Jan 2026
Specialization: Engineering of Machine Learning Systems, Trustworthy ML, Big Data Platforms. - Completed 30 ECTS of rigorous coursework focusing on the architecture and reliability of AI systems.	
Universidad San Francisco de Quito (USFQ) <i>Bachelor of Engineering, Computer Science</i>	Quito, Ecuador Expected Dec 2026

TECHNICAL SKILLS

Languages: Python, SQL, Bash
AI & ML: Generative AI, LLMs, RAG, LLM Evaluation, Forecasting, ML Systems
Infrastructure: Snowflake, Mage AI, Docker, CI/CD (GitHub Actions), Git
Concepts: MLOps, Data Pipelines, Reproducibility, Validation, CRISP-DM

AWARDS & CERTIFICATIONS

2nd Place, IEEEExtreme 18.0 (Ecuador): Global 24-hour algorithmic programming competition.
Technical AI Safety Course (BlueDot Impact): AI alignment curriculum.
IBM Machine Learning Specialization: Professional Certificate (Coursera).